

Sustainability aware asset management

Lecture 2: Expected Return and Plug-in Estimator

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Objectives of the Lecture

In the first lecture, we have studied static, single-period asset allocation decisions, assuming expected returns and the covariance matrix are known.

In practice, asset allocation is based on some estimates or predictions of expected returns and covariance matrix (inputs).

In this lecture, we discuss:

- how inputs should be estimated
- how benchmarking changes optimal weights but not required inputs
- what are the properties of the **plug-in estimator** of portfolio weights
 - Static properties
 - Dynamic properties

- **Dahlquist and Harvey** (2001), *Global Tactical Asset Allocation*, Working Paper.

⇒ Description of the three levels of asset allocation

- **Jobson and Korkie** (1980), *Estimation for Markowitz Efficient Portfolios*, Journal of the American Statistical Association, 75(371), 544–554.

- **DeMiguel and Nogales** (2009), *Portfolio Selection with Robust Estimation*, Operations Research, 57(3), 560–577

⇒ Illustration of the estimation error problem.

Objectives of the lecture

- **Estimating Expected Returns**
- Benchmarking vs Active Management
- Properties of Portfolio Weights

Estimating Risk Premia – Unconditional Mean

Given historical prices (including dividends) $\{P_0, \dots, P_T\}$, we compute returns

$$R_{t+1} = \frac{P_{t+1} + D_{t+1} - P_t}{P_t} \quad t = 0, \dots, T-1$$

Using the risk-free rate, $R_{f,t}$, we obtain excess returns:

$$R_{t+1}^e = R_{t+1} - R_{f,t}.$$

We then estimate unconditional risk premia as:

$$\hat{\mu}^e = \frac{1}{T} \sum_{t=1}^T R_t^e$$

Properties:

- easy to compute and update
- assumes constant risk premia
- with i.i.d. normal returns, the central limit theorem states that $\sqrt{T} (\hat{\mu}^e - \mu^e) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$

Estimating Risk Premia – Unconditional Mean

Example: Assume $\hat{\mu}^e = 6\%$, $\sigma = 15\%$ and $T = 120$ (10 years of monthly data).

The standard error of the sample mean is

$$\text{SE}(\hat{\mu}^e) = \frac{\sigma}{\sqrt{T}} = \frac{0.15}{\sqrt{120}} \approx 0.0137 \text{ (i.e., 1.37\%)}.$$

This implies a 95% confidence interval of

$$\mu^e \pm 1.96 \text{SE}(\hat{\mu}^e) \approx 6\% \pm 2.685\% \quad \Rightarrow \quad [3.31\%, 8.69\%].$$

Risk premia are difficult to estimate precisely

Remark: Most models are designed for daily returns. However, the asset allocation is often performed at the weekly or monthly frequency ($\rightarrow$ Temporal Aggregation).

Expected Returns – Predictability

Predicting equity returns is a difficult task:

- Returns are highly volatile
- Markets are quite efficient in incorporating information

The value of the firm is the discounted value of the cash flows

$$P_t = \mathbb{E}_t \sum_{i=0}^{\infty} \frac{CF_{t+i}}{(1 + R_{t+i}^D)^i} \quad \text{with} \quad R_{t+i}^D = R_{f,t} + \beta_i(R_{m,t+i} - R_{f,t})$$

There are at least four fundamental sources of predictability in equity returns:

- Expected cash flows (numerator)
- Risk-free interest rate (denominator)
- Expected firm's risk exposure (denominator)
- Expected market risk premium (denominator)

Expected Returns – Predictability

- **Expected cash flows:** They often move with the business cycle. Both are slow-moving and persistent. Short and long horizon predictability.
- **Risk-free rate:** It mostly reflects expectations about monetary policy.
- **Expected firm's risk exposure:** It depends on the industry and on the firm's capital structure (leverage). Its changes are slow, persistent, and partly predictable.
- **Expected market risk premium:** It is also related to the business cycle. During recessions, the premium must be high to attract investors into the market.

A common component of all four fundamental variables is the **business cycle**. The business cycle is persistent, and it is possible to partially predict real economic growth:

- Consumers tend to smooth consumption.
- Investment is sticky, i.e., corporate investment in projects is usually long-term.
- Government expenditures have a low level of variability.

Estimating Risk Premia: Linear Conditional Mean

Conditional risk premia for next period can be defined as linear functions of current macro-economic variables or firm-specific variables:

Time series:

$$R_{t+1}^e = \alpha + \beta' \mathbf{z}_t + \varepsilon_{t+1}, \quad t = 1, \dots, T,$$

where $\mathbf{z}_t$ may include business cycle indicators, such as interest rate, inflation rate, or dividend yield.

Cross section:

$$R_{i,t+1}^e = \alpha + \beta' \mathbf{x}_{i,t} + \varepsilon_{i,t+1}, \quad t = 1, \dots, T, \quad i = 1, \dots, N,$$

where $\mathbf{x}_{i,t}$ may include the market beta.

Remark: For long-term predictions, we can use a Vector AutoRegression (VAR) Model, which allows us to predict the predictors.

Theory

- Market portfolio (CAPM)
- I-CAPM: Intertemporal hedge portfolios (portfolios maximally correlated with changes in investment opportunities)

Macroeconomic factors

- Default premium
- Term premium
- Industrial production
- Inflation

Fundamental factors (accounting-based)

- Size
- Book to market
- Earnings to market
- Cash-flow to market
- Dividend yield
- Industry factors

Statistical procedure

- Factor analysis
- Principal components

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Taming the Factor Zoo: A Test of New Factors

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The factor zoo contains 98 tradable and 1 nontradable (No. 97) factors for monthly data from July 1980 to December 2016. In addition to these publicly available factors, we follow Fama and French (1993) to construct value-weighted portfolios as factors using firm characteristics collected in Green et al. (2016). For each characteristic therein, we sort all stocks into deciles based on their previous year-end values, then build and rebalance a long-short portfolio (top 30% - bottom 30% or 1-0 dummy difference) every June. For factor classification, “M” is Momentum, “V” is Value-versus-Growth, “I” is Investment, “P” is Profitability, “IN” is Intangibles, and “T” is Trading Frictions.

Factor Zoo

ID	Description	Year.pub	Year.end	Avg.Ret.	Ann.SR	Reference
1	Excess Market Return	1972	1965	0.64%	50.6%	Jensen, Black, and Scholes (1972)
2	Market Beta	1973	1968	-0.08%	-5.4%	Fama and MacBeth (1973)
3	Earnings to price	1977	1971	0.28%	29.7%	Basu (1977)
4	Dividend to price	1979	1977	0.01%	0.6%	Litzenberger and Ramaswamy (1979)
5	Unexpected quarterly earnings	1982	1980	0.12%	26.3%	Rendleman, Jones, and Latane (1982)
6	Share price	1982	1978	0.02%	2.2%	Miller and Scholes (1982)
7	Long-Term Reversal	1985	1982	0.34%	36.3%	Bondt and Thaler (1985)
8	Leverage	1988	1981	0.21%	24.3%	Bhandari (1988)
9	Cash flow to debt	1989	1984	-0.09%	-17.0%	Ou and Penman (1989)
10	Current ratio	1989	1984	0.06%	7.7%	Ou and Penman (1989)
11	% change in current ratio	1989	1984	0.00%	0.5%	Ou and Penman (1989)
12	% change in quick ratio	1989	1984	-0.04%	-11.9%	Ou and Penman (1989)
13	% change sales-to-inventory	1989	1984	0.17%	46.2%	Ou and Penman (1989)
14	Quick ratio	1989	1984	-0.02%	-2.9%	Ou and Penman (1989)
15	Sales to cash	1989	1984	0.01%	1.5%	Ou and Penman (1989)
16	Sales to inventory	1989	1984	0.09%	16.1%	Ou and Penman (1989)
17	Sales to receivables	1989	1984	0.14%	22.8%	Ou and Penman (1989)
18	Bid-ask spread	1989	1979	-0.04%	-3.3%	Amihud and Mendelson (1989)
19	Depreciation/PP&E	1992	1988	0.11%	12.1%	Holthausen and Larcker (1992)
20	% change in depreciation	1992	1988	0.08%	23.1%	Holthausen and Larcker (1992)
21	Small Minus Big	1993	1991	0.21%	24.5%	Fama and French (1993)
22	High Minus Low	1993	1991	0.28%	34.3%	Fama and French (1993)
23	Short-Term Reversal	1993	1989	0.15%	21.7%	Jegadeesh and Titman (1993)
24	6-month momentum	1993	1989	0.21%	27.8%	Jegadeesh and Titman (1993)

Factor Zoo

ID	Description	Year.pub	Year.end	Avg.Ret.	Ann.SR	Reference
25	36-month momentum	1993	1989	0.09%	13.4%	Jegadeesh and Titman (1993)
26	Sales growth	1994	1990	0.04%	5.8%	Lakonishok, Shleifer, and Vishny (1994)
27	Cash flow-to-price	1994	1990	0.31%	32.5%	Lakonishok, Shleifer, and Vishny (1994)
28	New equity issue	1995	1990	0.10%	8.7%	Loughran and Ritter (1995)
29	Dividend initiation	1995	1988	-0.03%	-3.4%	Michaely, Thaler, and Womack (1995)
30	Dividend omission	1995	1988	-0.18%	-18.0%	Michaely, Thaler, and Womack (1995)
31	Working capital accruals	1996	1991	0.22%	46.0%	Sloan (1996)
32	Sales to price	1996	1991	0.35%	41.8%	Barbee Jr, Mukherji, and Raines (1996)
33	Capital turnover	1996	1993	-0.11%	-16.6%	Haugen and Baker (1996)
34	Momentum	1997	1993	0.63%	50.2%	Carhart (1997)
35	Share turnover	1998	1991	-0.02%	-2.1%	Datar, Naik, and Radcliffe (1998)
36	% change in gross margin—% change in sales	1998	1988	-0.05%	-12.4%	Abarbanell and Bushee (1998)
37	% change in sales—% change in inventory	1998	1988	0.14%	42.1%	Abarbanell and Bushee (1998)
38	% change in sales—% change in A/R	1998	1988	0.14%	43.5%	Abarbanell and Bushee (1998)
39	% change in sales—% change in SG&A	1998	1988	0.09%	19.6%	Abarbanell and Bushee (1998)
40	Effective Tax Rate	1998	1988	-0.04%	-9.1%	Abarbanell and Bushee (1998)
41	Labor Force Efficiency	1998	1988	-0.03%	-8.5%	Abarbanell and Bushee (1998)
42	Ohlson's O-score	1998	1995	0.05%	9.3%	Dichev (1998)
43	Altman's Z-score	1998	1995	0.20%	22.1%	Dichev (1998)
44	Industry adjusted % change in capital expenditures	1998	1988	0.10%	20.5%	Abarbanell and Bushee (1998)
45	Number of earnings increases	1999	1992	0.01%	2.8%	Barth, Elliott, and Finn (1999)
46	Industry momentum	1999	1995	0.01%	1.4%	Moskowitz and Grinblatt (1999)
47	Financial statements score	2000	1996	0.08%	18.4%	Piotroski (2000)
48	Industry-adjusted book to market	2000	1998	0.22%	38.0%	Asness, Porter, and Stevens (2000)
49	Industry-adjusted cash flow to price ratio	2000	1998	0.26%	52.1%	Asness, Porter, and Stevens (2000)
50	Industry-adjusted change in employees	2000	1998	-0.01%	-1.5%	Asness, Porter, and Stevens (2000)
51	Industry-adjusted size	2000	1998	0.36%	36.3%	Asness, Porter, and Stevens (2000)
52	Dollar trading volume	2001	1995	0.38%	35.8%	Chordia, Subrahmanyam, and Anshuman (2001)
53	Volatility of liquidity (dollar trading volume)	2001	1995	0.20%	38.8%	Chordia, Subrahmanyam, and Anshuman (2001)
54	Volatility of liquidity (share turnover)	2001	1995	0.02%	2.1%	Chordia, Subrahmanyam, and Anshuman (2001)

Factor Zoo

ID	Description	Year.pub	Year.end	Avg.Ret.	Ann.SR	Reference
55	Advertising Expense-to-market	2001	1995	-0.13%	-15.6%	Chan, Lakonishok, and Sougiannis (2001)
56	R&D Expense-to-market	2001	1995	0.34%	36.2%	Chan, Lakonishok, and Sougiannis (2001)
57	R&D-to-sales	2001	1995	0.06%	5.5%	Chan, Lakonishok, and Sougiannis (2001)
58	Kaplan-Zingales Index	2001	1997	0.22%	25.3%	Lamont, Polk, and Saaá-Requejo (2001)
59	Change in inventory	2002	1997	0.18%	40.7%	Thomas and Zhang (2002)
60	Change in tax expense	2002	1997	0.09%	18.0%	Thomas and Zhang (2002)
61	Illiquidity	2002	1997	0.34%	28.6%	Amihud (2002)
62	Liquidity	2003	2000	0.38%	38.6%	Pástor and Stambaugh (2003)
63	Idiosyncratic return volatility	2003	1997	0.07%	5.1%	Ali, Hwang, and Trombley (2003)
64	Growth in long term net operating assets	2003	1993	0.22%	51.8%	Fairfield, Whisenant, and Yohn (2003)
65	Order backlog	2003	1999	0.05%	5.7%	Rajgopal, Shevlin, and Venkatachalam (2003)
66	Changes in Long-term Net Operating Assets	2003	1993	0.24%	56.0%	Fairfield, Whisenant, and Yohn (2003)
67	Cash flow to price ratio	2004	1997	0.27%	31.7%	Desai, Rajgopal, and Venkatachalam (2004)
68	R&D increase	2004	2001	0.06%	11.1%	Eberhart, Maxwell, and Siddique (2004)
69	Corporate investment	2004	1995	0.13%	36.4%	Titman, Wei, and Xie (2004)
70	Earnings volatility	2004	2001	0.10%	10.7%	Francis et al. (2004)
71	Abnormal Corporate Investment	2004	1995	0.13%	31.2%	Titman, Wei, and Xie (2004)
72	Net Operating Assets	2004	2002	0.31%	66.6%	Hirshleifer et al. (2004)
73	Changes in Net Operating Assets	2004	2002	0.14%	41.6%	Hirshleifer et al. (2004)
74	Tax income to book income	2004	2000	0.14%	28.3%	Lev and Nissim (2004)
75	Price delay	2005	2001	0.07%	16.8%	Hou and Moskowitz (2005)
76	# Years since first Compustat coverage	2005	2001	0.01%	1.1%	Jiang, Lee, and Zhang (2005)
77	Growth in common shareholder equity	2005	2001	0.15%	27.6%	Richardson et al. (2005)
78	Growth in long-term debt	2005	2001	0.06%	13.3%	Richardson et al. (2005)
79	Change in Current Operating Assets	2005	2001	0.19%	34.6%	Richardson et al. (2005)
80	Change in Current Operating Liabilities	2005	2001	0.03%	6.3%	Richardson et al. (2005)
81	Changes in Net Noncash Working Capital	2005	2001	0.11%	25.2%	Richardson et al. (2005)
82	Change in Noncurrent Operating Assets	2005	2001	0.21%	44.5%	Richardson et al. (2005)
83	Change in Noncurrent Operating Liabilities	2005	2001	0.04%	9.6%	Richardson et al. (2005)
84	Change in Net Noncurrent Operating Assets	2005	2001	0.23%	35.4%	Richardson et al. (2005)

Factor Zoo

ID	Description	Year.pub	Year.end	Avg.Ret.	Ann.SR	Reference
85	Change in Net Financial Assets	2005	2001	0.23%	59.0%	Richardson et al. (2005)
86	Total accruals	2005	2001	0.19%	44.8%	Richardson et al. (2005)
87	Change in Short-term Investments	2005	2001	-0.03%	-8.3%	Richardson et al. (2005)
88	Change in Financial Liabilities	2005	2001	0.18%	56.1%	Richardson et al. (2005)
89	Change in Book Equity	2005	2001	0.17%	30.0%	Richardson et al. (2005)
90	Financial statements performance	2005	2001	0.17%	37.1%	Mohanram (2005)
91	Change in 6-month momentum	2006	2006	0.21%	29.8%	Gettleman and Marks (2006)
92	Growth in capital expenditures	2006	1999	0.14%	30.4%	Anderson and Garcia-Feijoo (2006)
93	Return volatility	2006	2000	-0.02%	-1.7%	Ang et al. (2006)
94	Zero trading days	2006	2003	-0.05%	-4.4%	Liu (2006)
95	Three-year Investment Growth	2006	1999	0.11%	23.6%	Anderson and Garcia-Feijoo (2006)
96	Composite Equity Issuance	2006	2003	-0.01%	-2.2%	Daniel and Titman (2006)
97	Net equity finance	2006	2000	0.08%	9.7%	Bradshaw, Richardson, and Sloan (2006)
98	Net debt finance	2006	2000	0.17%	48.3%	Bradshaw, Richardson, and Sloan (2006)
99	Net external finance	2006	2000	0.22%	38.6%	Bradshaw, Richardson, and Sloan (2006)
100	Revenue Surprises	2006	2003	0.05%	9.0%	Jegadeesh and Livnat (2006)
101	Industry Concentration	2006	2001	0.03%	3.8%	Hou and Robinson (2006)
102	Whited-Wu Index	2006	2001	-0.02%	-2.6%	Whited and Wu (2006)
103	Return on invested capital	2007	2005	0.18%	29.3%	Brown and Rowe (2007)
104	Debt capacity/firm tangibility	2007	2000	0.05%	7.1%	Almeida and Campello (2007)
105	Payout yield	2007	2003	0.16%	17.5%	Boudoukh et al. (2007)
106	Net payout yield	2007	2003	0.16%	17.2%	Boudoukh et al. (2007)
107	Net debt-to-price	2007	1950	0.02%	2.5%	Penman, Richardson, and Tuna (2007)
108	Enterprise book-to-price	2007	2001	0.14%	14.7%	Penman, Richardson, and Tuna (2007)
109	Change in shares outstanding	2008	1969	0.24%	36.1%	Pontiff and Woodgate (2008)
110	Abnormal earnings announcement volume	2008	2006	-0.08%	-17.0%	Lerman, Livnat, and Mendenhall (2008)
111	Earnings announcement return	2008	2004	0.02%	6.8%	Brandt et al. (2008)
112	Seasonality	2008	2002	0.16%	17.3%	Heston and Sadka (2008)
113	Changes in PPE and Inventory-to-assets	2008	2005	0.19%	42.0%	Lyandres, Sun, and Zhang (2008)
114	Investment Growth	2008	2003	0.17%	39.5%	Xing (2008)
115	Composite Debt Issuance	2008	2005	0.08%	21.6%	Lyandres, Sun, and Zhang (2008)

Factor Zoo

ID	Description	Year.pub	Year.end	Avg.Ret.	Ann.SR	Reference
116	Return on net operating assets	2008	2002	0.09%	8.6%	Soliman (2008)
117	Profit margin	2008	2002	0.02%	4.4%	Soliman (2008)
118	Asset turnover	2008	2002	0.06%	6.7%	Soliman (2008)
119	Industry-adjusted change in asset turnover	2008	2002	0.14%	41.1%	Soliman (2008)
120	Industry-adjusted change in profit margin	2008	2002	-0.01%	-3.2%	Soliman (2008)
121	Cash productivity	2009	2009	0.27%	37.6%	Chandrashekar and Rao (2009)
122	Sin stocks	2009	2006	0.44%	41.6%	Hong and Kacperczyk (2009)
123	Revenue surprise	2009	2005	0.12%	19.3%	Kama (2009)
124	Cash flow volatility	2009	2008	0.20%	26.6%	Huang (2009)
125	Absolute accruals	2010	2008	-0.05%	-8.6%	Bandyopadhyay, Huang, and Wirjanto (2010)
126	Capital expenditures and inventory	2010	2006	0.19%	42.8%	Chen and Zhang (2010)
127	Return on assets	2010	2005	-0.09%	-13.9%	Balakrishnan, Bartov, and Faurel (2010)
128	Accrual volatility	2010	2008	0.19%	26.6%	Bandyopadhyay, Huang, and Wirjanto (2010)
129	Industry-adjusted Real Estate Ratio	2010	2005	0.11%	17.3%	Tuzel (2010)
130	Percent accruals	2011	2008	0.16%	35.0%	Hafzalla et al. (2011)
131	Maximum daily return	2011	2005	0.00%	-0.3%	Bali, Cakici, and Whitelaw (2011)
132	Operating Leverage	2011	2008	0.20%	32.8%	Novy-Marx (2011)
133	Inventory Growth	2011	2009	0.13%	30.1%	Belo and Lin (2011)
134	Percent Operating Accruals	2011	2008	0.15%	28.9%	Hafzalla et al. (2011)
135	Enterprise multiple	2011	2009	0.11%	17.6%	Loughran and Wellman (2011)
136	Cash holdings	2012	2009	0.13%	15.3%	Palazzo (2012)
137	HML Devil	2013	2011	0.23%	22.6%	Asness and Frazzini (2013)
138	Gross profitability	2013	2010	0.15%	22.5%	Novy-Marx (2013)
139	Organizational Capital	2013	2008	0.21%	31.9%	Eisfeldt and Papanikolaou (2013)
140	Betting Against Beta	2014	2012	0.91%	92.8%	Frazzini and Pedersen (2014)
141	Quality Minus Junk	2014	2012	0.43%	60.1%	Asness, Frazzini, and Pedersen (2019)
142	Employee growth rate	2014	2010	0.08%	12.9%	Belo, Lin, and Bazdresch (2014)
143	Growth in advertising expense	2014	2010	0.07%	13.0%	Lou (2014)
144	Book Asset Liquidity	2014	2006	0.09%	12.3%	Ortiz-Molina and Phillips (2014)
145	Robust Minus Weak	2015	2013	0.34%	49.8%	Fama and French (2015)
146	Conservative Minus Aggressive	2015	2013	0.26%	46.8%	Fama and French (2015)
147	HXZ Investment	2015	2012	0.34%	64.7%	Hou, Xue, and Zhang (2015)
148	HXZ Profitability	2015	2012	0.57%	77.5%	Hou, Xue, and Zhang (2015)
149	Intermediary Investment	2016	2012			He, Kelly, and Manela (2017)
150	Convertible debt indicator	2016	2012	0.11%	26.4%	Valta (2016)

Guidelines for Backtesting

- Use theory and intuition to guide model specification
- Constrain specification search process before confronting data
 - explicitly list all model specifications to be considered
 - single-shot model comparison
 - stop regardless of outcome
- Check if results are reasonable
 - 10% of realized information ratios
 - information ratios likely due to data mining or mistakes
- Careful out-of-sample analysis

Estimating Covariance Matrix – Unconditional Moment

Given excess returns, $\{\mathbf{R}_t^e\}_{t=1}^T$, the sample covariance matrix can be estimated as:

$$\hat{\Sigma} = \frac{1}{T-1} \sum_{t=1}^T (\mathbf{R}_t^e - \hat{\mu}^e)(\mathbf{R}_t^e - \hat{\mu}^e)'.$$

Properties:

- easy to compute and update
- assumes constant return distribution

However:

- the horizon of the allocation is not necessarily the same as the frequency of the data
- we know that volatilities and correlations are time varying.

Estimating the Covariance Matrix – Rolling Window

An alternative approach that accounts for time-variability is the use of **Rolling Windows**.

- Consider the first τ observations ($\tau \ll T$).
- Estimate risk premia ($\hat{\mu}_\tau$) and the covariance matrix ($\hat{\Sigma}_\tau$) using $t = 1, \dots, \tau$.

$$\hat{\mu}_\tau^e = \frac{1}{\tau} \sum_{t=1}^{\tau} \mathbf{R}_t^e \quad \Rightarrow \quad \hat{\Sigma}_\tau = \frac{1}{\tau - 1} \sum_{t=1}^{\tau} (\mathbf{R}_t^e - \hat{\mu}_\tau^e)(\mathbf{R}_t^e - \hat{\mu}_\tau^e)'$$

- Roll the sample by 1 observation.
- Estimate risk premia ($\hat{\mu}_{\tau+1}$) and the covariance matrix ($\hat{\Sigma}_{\tau+1}$) using $t = 2, \dots, \tau + 1$
- ...
- Estimate risk premia ($\hat{\mu}_T$) and the covariance matrix ($\hat{\Sigma}_T$) using $t = T - \tau + 1, \dots, T$

Estimating the Covariance Matrix – EWMA

A slightly better estimator is the **Exponentially Weighted Moving Average (EWMA)**.

Consider a memory parameter φ ($0 < \varphi < 1$), an initial risk premium $\hat{\mu}_0^e$, and an initial covariance matrix $\hat{\Sigma}_0$. $\hat{\mu}_0^e$ and $\hat{\Sigma}_0$ can be based on a pre-sample.

At date t , compute the risk premium and the covariance matrix as:

$$\begin{aligned}\hat{\mu}_t^e &= \varphi \hat{\mu}_{t-1}^e + (1 - \varphi) \mathbf{R}_t^e, \\ \hat{\Sigma}_t &= \varphi \hat{\Sigma}_{t-1} + (1 - \varphi) (\mathbf{R}_t^e - \hat{\mu}_{t-1}^e)(\mathbf{R}_t^e - \hat{\mu}_{t-1}^e)'.\end{aligned}$$

Remark: Problems related to the number of observations are even more severe.

Uncertainty in the Covariance Matrix

For N assets, the number of parameters to estimate is $N(N + 1)/2$, the number of observations is NT .

For $N = 500$ stocks and $T = 600$ months of data (50 years):

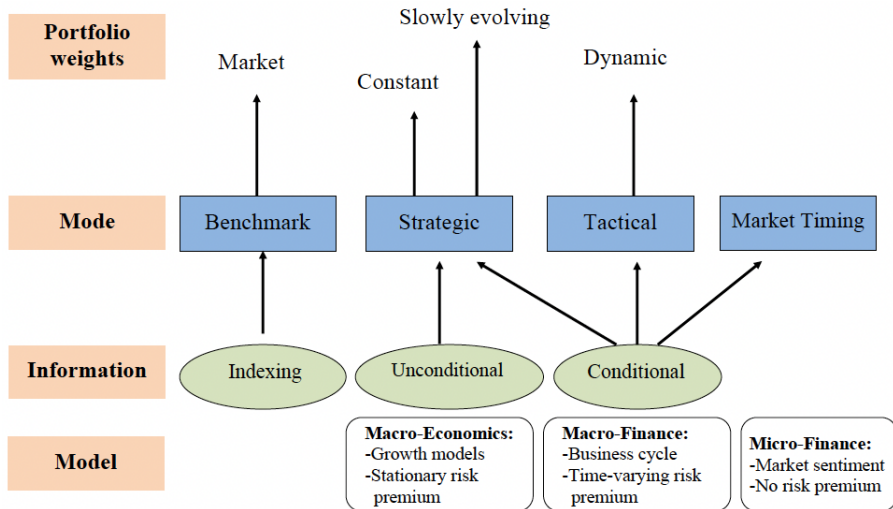
- $N(N + 1)/2 = 125,250$ unique parameters
- $NT = 300,000$ observations
- so each parameter is estimated with 2.4 observations on average.

The sample covariance matrix is singular when $N > T - 1$.

Objectives of the lecture

- Estimating Expected Returns
- **Benchmarking vs Active Management**
- Properties of Portfolio Weights

Expected Returns



Expected Returns – Using Conditional Information

- **Benchmark allocation:** No conditioning information is used.
- **Strategic allocation:** Long-horizon forecasting model is used to predict 5- or 10-year returns.
- **Tactical allocation:** Conditioning information is always used. Expected short-term returns, standard deviations, and correlations are the basis for weight changes.
- **Unconditional approach:** Expected returns and the covariance matrix are constant over time. It assumes that historical average returns are good predictors of future returns.
- **Conditional approach:** We use information available today. The mean changes through time depending on the values of the information variables.

Active versus Passive Management

Most asset managers are evaluated based on running the adjusted CAPM regression:

$$R_{p,t+1} - R_{f,t} = \alpha_p + \beta_p (R_{b,t+1} - R_{f,t}) + \varepsilon_{p,t+1}.$$

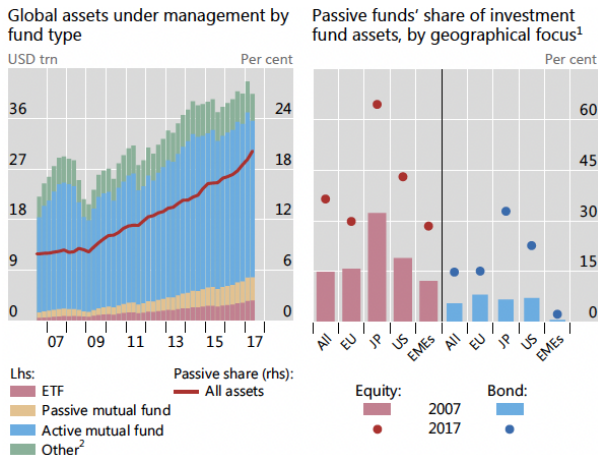
where $R_{p,t+1}$ is the fund return and $R_{b,t+1}$ is the return of the benchmark of the fund

Passive management $\approx$ beta exposure	versus	active management $\approx$ alpha
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We distinguish **three levels of asset allocation** (Dahlquist and Harvey, 2001):

- **Benchmark / Passive management** (index funds, passive mutual funds)
- **Strategic / Active management** (active mutual funds, long-term managers: pension funds, insurers, sovereign wealth funds)
- **Tactical / High-frequency trading** (hedge funds)

Relative importance of passive and active management – AUM



Note: 1 As of end-June for each year. 2 Includes investment fund assets of closed-end funds, hedge funds, insurance funds, investment trusts and pension funds. Source: BIS Quarterly Review, March 2018, The implications of passive investing for securities markets

Relative importance of passive and active management – AuM USD tln



Source:

<https://www.wilmingtontrust.com/library/article/revisiting-active-passive-investing-debate>

Benchmark (Passive) Asset Allocation

Investors exactly replicate the investment weights of a benchmark index.

Example: If the benchmark is MSCI world, the benchmark allocation has the same weights as this index.

This type of asset allocation is sometimes referred to as indexing. No information is used other than the usual details of **indexing**: market weights, delistings, new listings, buybacks, secondary market offerings, dividends, and warrants.

If the benchmark is broad (MSCI world), this allocation is sometimes referred to as **Equilibrium Asset Allocation** because the CAPM implies that investors hold the world market portfolio in equilibrium, levered up or down to match desired risk aversion.

Remark: If we use historical covariance matrix, we can back out of the optimization the expected returns that exactly imply the market weights. These expected returns are often referred to as **Equilibrium Expected Returns**. (See Black-Litterman approach below)

Strategic Asset Allocation (SAA)

Investors with long-term horizon (say, 5 years) have a view that certain assets will out-perform or under-perform and adopt weights that deviate significantly from the index.

Example: The manager's view is that Japanese government bonds will underperform (relative to historical performance) over the next 5 years. She would underweight these bonds relative to the benchmark.

Even if weights are based on 5-year forecasts, it is common practice to update the 5-year forecasts every year and, hence, rebalance annually.

Deviation from the benchmark introduces **tracking error**, i.e., the standard deviation of the differences between the benchmark return and the portfolio return.

Strategic deviations from benchmark will induce a **'strategic' tracking error**.

Remark: SAA may be implemented using either static models or dynamic models.

Tactical Allocation (TAA)

Investors take short-term bets, usually 1 month to 1 quarter, and deviate from the strategic weights. This also induces tracking error. The difference between the strategic and tactical weights induces **'tactical' tracking error**.

Because of the frequent changes in investment weights in tactical programs, transactions costs are important. A common strategy is to minimize transactions costs by using futures or swaps (instead of stocks or bonds) to manage tactical deviations.

As futures and swaps can be used to manage bets at minimal cost, we could consider a fourth level of asset allocation – the **“High-Frequency Tactical Asset Allocation” (or Market Timing)**. It is run on a daily or intraday basis, using quantitative models.

Tactical Asset Allocation (TAA) is the approach to portfolio management of many hedge funds.

Short-term bets are placed for periods of typically less than a year, but it may also be used by an investment fund that simultaneously has strategic asset allocation objectives.

Portfolio Optimization with a Benchmark

- Passive investors directly invest in a pre-defined benchmark
- Active investors usually invest relative to a pre-defined benchmark
- A few active investors (hedge funds) invest with no pre-defined benchmark
- Allocation relative to a benchmark can have a long-term perspective (strategic allocation) or a short-term perspective (tactical allocation)
- Optimizing relative to a benchmark changes the optimization program but the required inputs remain the same

Tracking Error

The tracking error reflects the difference between the return on the strategic or tactical asset allocation and the return on the benchmark portfolio:

$$TE_t = \sqrt{\text{Var}_t(R_{p,t+1} - R_{b,t+1})}.$$

Rationales for Benchmark-Relative Investment:

- Risk perspective: A benchmark anchors the portfolio in risk-return space and thus gives clients confidence in what risk the investment carries.
- Return perspective: Some claim it is easier to forecast relative returns.
- More plausible: Estimation error is smaller for relative than for absolute return forecasts.

Active Risk/Return Perspective

In the asset management industry, managers are often rewarded according to their performance relative to a benchmark. So, managers are likely to focus on active return and risk, rather than on total return and risk of portfolio.

If only the tactical position matters, we can reformulate the optimization. Let α_b be the vector of weights of the benchmark and α_p the vector of weights of the portfolio. Let α_a be the vector of tactical bets (or **active portfolio bets**) on each asset in the portfolio, i.e., α_a measures the deviation from the benchmark:

$$\alpha_a = \alpha_p - \alpha_b.$$

The **tracking difference** is defined as: $e_{t+1} = \alpha_a' \mathbf{R}_{t+1}$

Now, the important quantities are

- The **Tracking Error**: $\text{TE}_t = \sqrt{\alpha_a' \Sigma_t \alpha_a}$ (Risk)
- The **expected return (Alpha)** relative to benchmark: $\mathbb{E}_t[e_{t+1}] = \alpha_a' \mu_t$ (Return)

Active Risk/Return Perspective

The objective of TAA is then to **minimize the benchmark tracking errors**, while at the same time outperforming the benchmark on average by a certain amount.

The problem to solve is:

$$\begin{cases} \max_{\{\alpha_a\}} & \mathbb{E}_t[e_{t+1}] - \frac{\lambda}{2} \text{Var}_t[e_{t+1}] = \alpha'_a \mu_t - \frac{\lambda}{2} \alpha'_a \Sigma_t \alpha_a \\ \text{s.t.} & \alpha'_a \mathbf{e} = 0 \quad (\text{sum equal to 0}) \end{cases}$$

Proposition: The weights of the **active portfolio bets** are:

$$\alpha_a^* = \frac{1}{\lambda} \Sigma_t^{-1} \left(\mu_t - \frac{\mathbf{e}' \Sigma_t^{-1} \mu_t}{\mathbf{e}' \Sigma_t^{-1} \mathbf{e}} \mathbf{e} \right) = \frac{1}{\lambda} \left(\frac{\Sigma_t^{-1} (\mu_t \mathbf{e}' - \mathbf{e} \mu_t') \Sigma_t^{-1}}{\mathbf{e}' \Sigma_t^{-1} \mathbf{e}} \right) \mathbf{e}.$$

Properties:

- If the benchmark portfolio is α_b , the total portfolio is $\alpha_p = \alpha_b + \alpha_a$.
- The information ratio is defined as $IR_{p,t} = \frac{\mathbb{E}_t[e_{t+1}]}{\text{Var}_t[e_{t+1}]} = \frac{\alpha'_a \mu_t}{\sqrt{\alpha'_a \Sigma_t \alpha_a}}.$

Objectives of the lecture

- Estimating Expected Returns
- Benchmarking vs Active Management
- **Properties of Portfolio Weights**

Plug-in Estimator under Parameter Uncertainty

In the case with a risk-free asset, the weights of the optimal mean-variance (tangency) portfolio are obtained assuming that expected returns μ^e and the covariance matrix Σ are known:

$$\alpha^* = \frac{1}{\lambda} \Sigma^{-1} \mu^e \quad \text{where} \quad \mu^e = \mathbf{R} - R_f \mathbf{e}$$

Properties:

- intuitive and computationally simple
- asymptotically consistent
- unbiased under normality

In practice, μ^e and Σ are estimated, yielding:

$$\hat{\alpha} = \frac{1}{\lambda} \hat{\Sigma}^{-1} \hat{\mu}^e \quad (\text{plug-in estimator}).$$

$\hat{\alpha}$ is estimated with uncertainty. Standard errors of $\hat{\alpha}$ can be approximated using a Taylor expansion and the standard errors of $\hat{\mu}^e$ and $\hat{\Sigma}$. This is easy to do with single risky asset, tedious with several risky assets.

Economic Loss from Parameter Uncertainty

Cho (2002), *Uncertainty in Second Moments: Implications for Portfolio Allocation*.

The effect of the uncertainty induced by the estimation of expected returns and the covariance matrix can be measured through certainty equivalent analysis.

The certainty equivalent is the value of the objective function evaluated at a given portfolio using the true parameters.

The (unattainable) optimal mean-variance portfolio (based on known expected returns and covariance matrix) gives:

$$\alpha^* = \frac{1}{\lambda} \Sigma^{-1} \mu^e \quad \text{and} \quad \text{CE} = \alpha^{*'} \mu^e - \frac{\lambda}{2} \alpha^{*'} \Sigma \alpha^*.$$

Plug-in portfolio weights and corresponding certainty equivalent (based on unknown expected returns and covariance matrix):

$$\hat{\alpha} = \frac{1}{\lambda} \hat{\Sigma}^{-1} \hat{\mu}^e \quad \text{and} \quad \widehat{\text{CE}} = \hat{\alpha}' \mu^e - \frac{\lambda}{2} \hat{\alpha}' \Sigma \hat{\alpha}.$$

Certainty Equivalent Loss

If we assume that $\hat{\alpha}$ is asymptotically unbiased, we have:

$$\mathbb{E}[\widehat{\text{CE}}] = \mathbb{E}[\hat{\alpha}]' \mu^e - \frac{\lambda}{2} \mathbb{E}[\hat{\alpha}' \Sigma \hat{\alpha}] \approx \alpha^{*'} \mu^e - \frac{\lambda}{2} \mathbb{E}[\hat{\alpha}' \Sigma \hat{\alpha}].$$

Certainty equivalent loss:

$$\begin{aligned} \text{CE}_{\text{loss}} &= \text{CE} - \mathbb{E}[\widehat{\text{CE}}] = \frac{\lambda}{2} (\mathbb{E}[\hat{\alpha}' \Sigma \hat{\alpha}] - \alpha^{*'} \Sigma \alpha^*) \\ &= \frac{\lambda}{2} \mathbb{E}[\text{tr}(\hat{\alpha}' \Sigma \hat{\alpha})] - \frac{\lambda}{2} \alpha^{*'} \Sigma \alpha^* = \frac{\lambda}{2} [\text{tr}(\mathbb{E}(\hat{\alpha} \hat{\alpha}' \Sigma))] - \frac{\lambda}{2} \alpha^{*'} \Sigma \alpha^* \\ &= \frac{\lambda}{2} \text{tr}(\text{Cov}[\hat{\alpha}] \Sigma) + \frac{\lambda}{2} \text{tr}(\mathbb{E}[\hat{\alpha}] \mathbb{E}[\hat{\alpha}]' \Sigma) - \frac{\lambda}{2} \alpha^{*'} \Sigma \alpha^* \\ &= \frac{\lambda}{2} \text{tr}(\text{Cov}[\hat{\alpha}] \Sigma) + \frac{\lambda}{2} \text{tr}(\alpha^* \alpha^{*'} \Sigma) - \frac{\lambda}{2} \alpha^{*'} \Sigma \alpha^* = \frac{\lambda}{2} \text{tr}(\text{Cov}[\hat{\alpha}] \Sigma) \end{aligned}$$

where the expectation is taken over the estimation errors (distribution of weights), not the future return distribution, which is already captured in CE.

Certainty Equivalent Loss

With a single risky asset (market portfolio), we have $\alpha^* = \frac{\mu_m^e}{\lambda \sigma_m^2}$ and $\text{CE}_{\text{loss}} = \frac{\lambda}{2} \text{Var}[\hat{\alpha}] \sigma_m^2$

Using a first-order Taylor expansion around true values (μ_m^e, σ_m^2) :

$$\hat{\alpha} = \frac{\hat{\mu}_m^e}{\lambda \hat{\sigma}_m^2} \approx \frac{\mu_m^e}{\lambda \sigma_m^2} + \frac{1}{\lambda \sigma_m^2} (\hat{\mu}_m^e - \mu_m^e) - \frac{\mu_m^e}{\lambda \sigma_m^4} (\hat{\sigma}_m^2 - \sigma_m^2) = \alpha^* + \frac{1}{\lambda \sigma_m^2} (\hat{\mu}_m^e - \mu_m^e) - \frac{\mu_m^e}{\lambda \sigma_m^4} (\hat{\sigma}_m^2 - \sigma_m^2)$$

Then, we find the estimator for the variance of $\hat{\alpha}$:

$$\widehat{\text{Var}}[\hat{\alpha}] = \frac{1}{\lambda^2} \widehat{\text{Var}} \left[\frac{\hat{\mu}_m^e}{\hat{\sigma}_m^2} \right] \approx \frac{\widehat{\text{Var}}[\hat{\mu}_m^e]}{\lambda^2 \hat{\sigma}_m^4} + \frac{(\hat{\mu}_m^e)^2 \widehat{\text{Var}}(\hat{\sigma}_m^2)}{\lambda^2 \hat{\sigma}_m^8} = \frac{1}{\lambda^2} \left(\frac{\hat{\mu}_m^e}{\hat{\sigma}_m^2} \right)^2 \left[\frac{\widehat{\text{Var}}[\hat{\mu}_m^e]}{(\hat{\mu}_m^e)^2} + \frac{\widehat{\text{Var}}(\hat{\sigma}_m^2)}{\hat{\sigma}_m^4} \right]$$

so that CE loss is estimated by: (SCV: squared coefficient of variation)

$$\text{CE}_{\text{loss}} = \frac{\lambda}{2} \widehat{\text{Var}}[\hat{\alpha}] \hat{\sigma}_m^2 = \frac{1}{2\lambda^2} \left(\frac{(\hat{\mu}_m^e)^2}{\hat{\sigma}_m^2} \right) [\text{SCV}[\hat{\mu}_m^e] + \text{SCV}[\hat{\sigma}_m^2]]$$

The first term in brackets corresponds to the contribution from estimating means, the second term to the contribution from estimating second moments.

Certainty Equivalent Loss – Example 1

Based on a single risky asset (CRSP value weighted portfolio) over 1963–2001. With a risk aversion parameter $\lambda = 2.13$, the optimal weight is $\alpha^* = 1$ and the corresponding certainty equivalent is $CE = 2.749\%$ per year.

Alternative models for estimating expected returns and the covariance matrix:

- Case 1: Returns are in fact iid normal
- Case 2: Returns are in fact drawn from a model with dynamic volatility.

(% per year)	Case 1: iid normal	Case 2: dynamic volatility
CE (optimal)	2.749%	2.749%
CE loss	0.588%	1.009%
Contribution from mean ($\hat{\mu}_m^e$)	0.576%	0.576%
Contribution from variance ($\hat{\sigma}_m^2$)	0.012%	0.430%

Comment: The loss due to failing to capture the dynamics of the second moment estimation can be as large as the loss due to the first moment estimation. This is the case when we estimate a constant covariance matrix while volatility vary over time.

Certainty Equivalent Loss – Example 2

Based on investing in 25 Fama-French portfolios over 1963–2001. Risk aversion parameter is $\lambda = 2.27$.

Alternative models for estimating expected returns and the covariance matrix:

- Case 1: Sample moments are the truth
- Case 2: Moments given by the Fama-French three-factor model are the truth.

(% per year)	Case 1: Sample	Case 2: FF 3-factor
CE (optimal)	80.6%	15.9%
CE loss	22.5%	4.0%
Contribution from mean ($\hat{\mu}^e$)	15.3%	1.8%
Contribution from variance ($\hat{\Sigma}$)	4.8%	1.6%

Comment: As the number of assets in the portfolio increases, the loss due to the uncertainty in second moments becomes larger, even if second moments remain constant. They are more accurately estimated but there are more parameters to estimate.

Illustration – Sensitivity to the expected return

Investment universe: 18 STOXX sectors in the Eurozone.

First case: Historical returns.

Revised case: +2%pts for BANK and MEDA, –2%pts for CNYL, others unchanged.

Weights determined via mean-variance optimization.

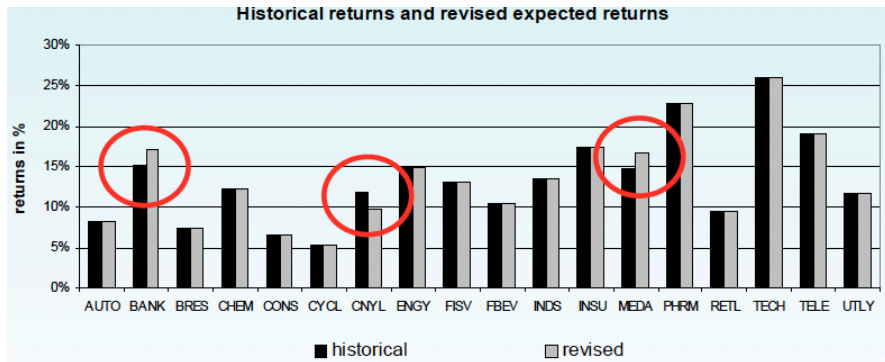
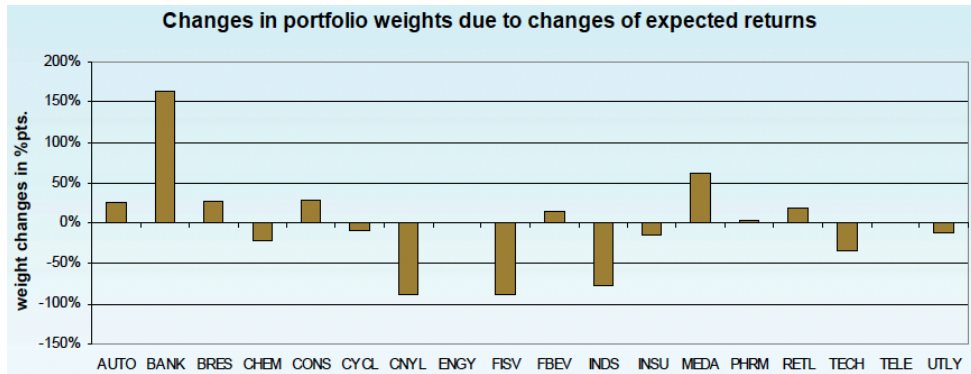


Illustration – Sensitivity to the expected return



Comments:

- Even small and selected changes in expected returns lead to huge unrealistic shifts in asset weights!
- Problems: Communication of results, (re-)allocation in real portfolios.

General conclusion: Plug-in estimates are extremely imprecise and, even in large sample, the asymptotic approximations are quite unreliable. Moreover, the precision of plug-in estimates deteriorates drastically with the number of assets.

Intuitively, this is because, as the number of assets increases, the number of unique elements of the return covariance matrix increases at a quadratic rate.

- **Jobson and Korkie (1980)** were among the first to document the finite-sample properties of plug-in estimates in a static setting.
- **DeMiguel and Nogales (2009)** extend their analysis to a dynamic setting.

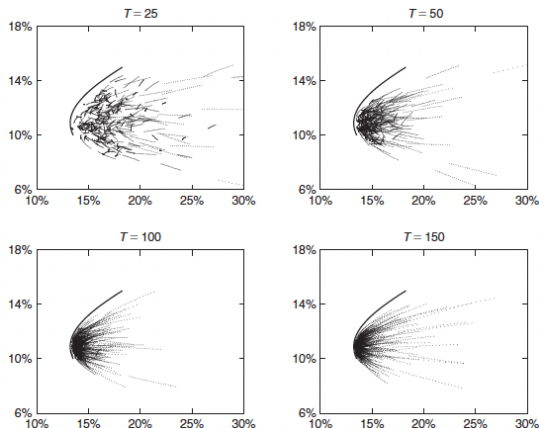
Jobson and Korkie Experiment – Static Allocation

Jobson and Korkie (1980), *Estimation for Markowitz Efficient Portfolios* (JASA)

Experiment:

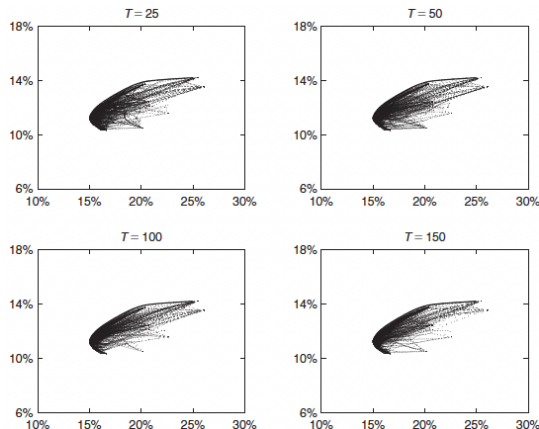
- ① Take 10 industry-sorted portfolios, assumed to be normally distributed
- ② Take the historical sample moments of the portfolios to be the “true moments”
- ③ Compute the true mean-variance efficient frontier (**asymptotic approximation**)
- ④ Simulate independent sets of 250 hypothetical return samples of different sample sizes from a normal distribution with the true moments
- ⑤ For each hypothetical sample, compute again plug-in estimates of the optimal weights along the mean–variance frontier
- ⑥ Evaluate the efficient frontier using the true moments.

Jobson and Korkie Experiment – Unconstrained mean-variance frontier



Note: The solid line in each plot is the unconstrained mean–variance frontier for 10 industry portfolios, taking sample moments as the truth. The dotted lines show the mean–variance trade-off, evaluated using the true moments, of 250 independent plug-in estimates for 25, 50, 100, and 150 simulated returns. Source: Brandt (2010)

Jobson and Korkie Experiment – Constrained mean-variance frontier



Note: The solid line in each plot is the constrained (nonnegative portfolio weights) mean-variance frontier for 10 industry portfolios, taking sample moments as the truth. The dotted lines show the mean-variance trade-off, evaluated using the true moments, of 250 independent plug-in estimates for 25, 50, 100, and 150 simulated returns. Source: Brandt (2010)

Jobson and Korkie Experiment: Static Allocation

Results:

- Mean-variance efficient portfolio weights are extremely noisy and on average considerably inferior to the true mean–variance frontier.
- Increasing the sample size, for example from 50 to 150, does not substantially reduce the sampling variability of the plug-in estimates.
- Asymptotic approximations are unreliable
- Constraining portfolio weights help reduce the sampling error, but clearly not to a point where one can trust the plug-in estimates, even for a sample as large as 150 months (more than 10 years of data).
- Plug-in estimates tend to exhibit extreme portfolio weights, which contradicts the notion diversification
- Small changes in the inputs (the risk premia and covariance matrix) result in large changes in the plug-in estimates.

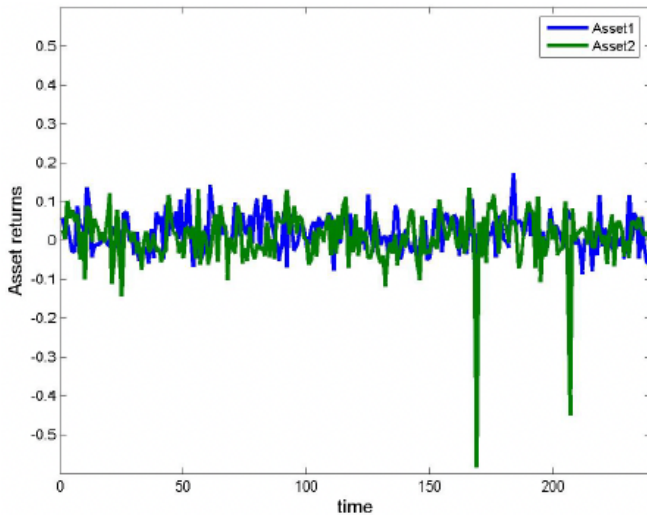
DeMiguel and Nogales Experiment – Dynamic Allocation

DeMiguel and Nogales (2009), *Portfolio Selection with Robust Estimation* (Operations Research)

Experiment:

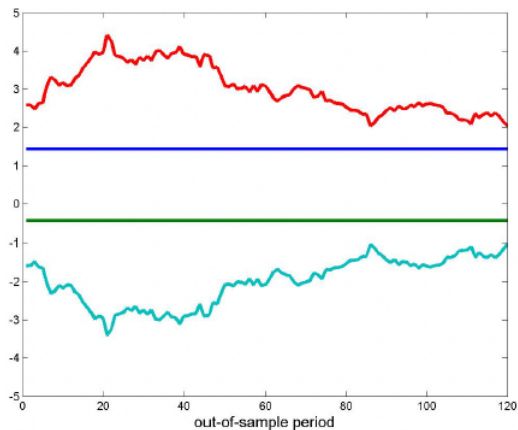
- 1 Take 2 assets with monthly i.i.d. returns drawn as $99\% \times N(\boldsymbol{\mu}, \boldsymbol{\Sigma}) + 1\% \times N(\boldsymbol{\mu}_d, \boldsymbol{\Sigma})$ with $\boldsymbol{\mu} = \begin{pmatrix} 0.01 \\ 0.01 \end{pmatrix}$, $\boldsymbol{\mu}_d = \begin{pmatrix} 0.01 \\ -0.5 \end{pmatrix}$, $\boldsymbol{\Sigma} = \begin{pmatrix} 0.0021 & 0 \\ 0 & 0.0021 \end{pmatrix}$ corresponding to an annual mean of 12% and an annual standard deviation of 16% in normal time.
- 2 From this distribution, simulate 240 hypothetical monthly returns
- 3 The first 120 returns are used to estimate the sample mean and covariance matrix
- 4 Compute the minimum-variance portfolio and the mean-variance portfolio for $\lambda = 1$
- 5 Then the sample window is shifted by 1 and the optimal portfolio is calculated
- 6 Repeat this procedure to obtain 120 different portfolios.

DeMiguel and Nogales Experiment – Time series of asset returns

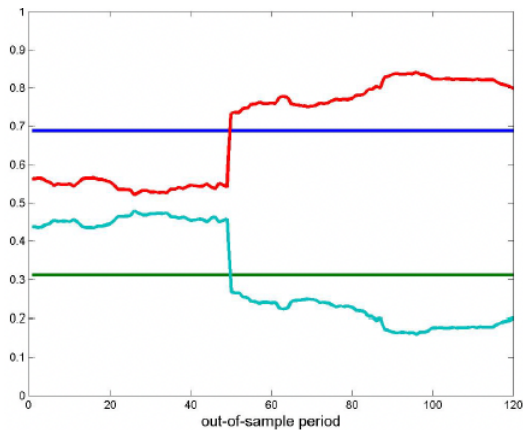


DeMiguel and Nogales Experiment: Dynamic Allocation

Mean-variance portfolio weights
($\lambda = 1$)



Minimum-variance portfolio weights
($\lambda \rightarrow \infty$)



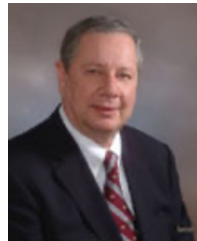
Straight lines: true portfolio weights (blue: asset 1, green: asset 2)

DeMiguel and Nogales Experiment – Results

- Mean-variance portfolios ($\lambda = 1$) are not well diversified: true weight of 143% for Asset 1 (range between 200% and 425%) and -43% for Asset 2 (range between -325% and -100%).
- Minimum-variance portfolios are better diversified: true weight of 69% for Asset 1 (range between 50% and 85%) and 31% for Asset 2 (range between 15% and 50%). Weights are highly affected by the jumps in the estimation window.
- Minimum-variance portfolios are much more stable: the instability due to estimation error in mean returns is much larger than the instability due to estimation error in the covariance matrix.
- Minimum-variance portfolios underestimate the weight of Asset 1 when there are no jumps in the estimation window and overestimate it when there are jumps.
- When investors impose no constraints, asset weights in the optimized portfolios almost always imply large short positions in many assets.
- When short positions are ruled out, the models often prescribe corner solutions, with zero weights in many assets and large weights in small cap assets.

“Mean-variance optimizers are, in a fundamental sense, estimation-error maximizers”

(Michaud, 1989, *The Markowitz Optimization Enigma: Is Optimized Optimal?* Financial Analysts Journal)



Portfolio optimizers amplify the effect of estimation error:

- They **overweight** securities that have unusually **high** estimated **risk premia** and/or low estimated risk
- They **underweight** securities that have unusually **low** estimated **risk premia** and/or high estimated risk
- Extreme estimates are more likely to contain estimation error ($\neq$ notion of diversification)

Plug-in Estimator: Summary

The use of the sample mean and covariance matrix of returns affects the performances of the **mean-variance portfolio construction**, because of estimation error:

- The resulting portfolios are very unstable
- The portfolio weights fluctuate substantially over time
- The performances are very poor out of sample.

The main source of the problem is estimation error in expected returns:

“The estimation error in the sample mean is so large that nothing much is lost in ignoring the mean altogether” (Jagannathan and Ma, 2003).

An alternative approach is to estimate minimum-variance portfolios for a given expected return, as it does not require the estimate of the expected return. Therefore, it should perform better out of sample. However, it is still quite vulnerable to the impact of estimation error on the covariance matrix.

End of Lecture 2